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Pain-control strategies for blind office endometrial biopsy: a systematic review with inert-controlled meta-analysis

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Pain-control strategies for blind office endometrial biopsy: a systematic review with inert-controlled meta-analysis

Title Page

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Systematic review registration: Not prospectively registered. The protocol, analysis plan, extracted datasets, code, figures, and screening records were deposited in Zenodo before journal submission for transparency and reproducibility (Zenodo concept DOI: https://doi.org/10.5281/zenodo.20116870).

Abstract

Background: Blind office endometrial biopsy is common, but pain during sampling can limit acceptability. Prior reviews have often combined different procedures, comparators, and pain timepoints. We evaluated randomized evidence for pain-control strategies during blind outpatient or office-based endometrial biopsy or sampling.

Methods: For the primary estimand, we restricted meta-analysis to randomized comparisons against placebo, saline, placebo gel, distilled water, or other inert controls, defined as controls without intended analgesic activity, that reported extractable intra-procedural pain scores on a 0-10 compatible scale. We searched PubMed, CENTRAL, Embase, Web of Science, Scopus, Wanfang, VIP, ClinicalTrials.gov, and WHO ICTRP from inception to 9 May 2026, with update checks of PubMed and ClinicalTrials.gov on 12 June 2026. CNKI, SinoMed/CBM, and ChiCTR were checked supplementally. Random-effects mean differences were pooled. Risk of bias was assessed with RoB 2 and certainty with GRADE.

Results: The PRISMA-counted searches identified 2632 records; 619 duplicates were removed and 2013 records were screened. Eight randomized trials involving 944 participants were included in the primary meta-analysis. Pain-control interventions lowered average pain during biopsy compared with inert controls (mean difference, -1.87 points; 95% confidence interval, -2.52 to -1.22; I-squared = 86.7%). REML and Hartung-Knapp sensitivity analyses remained directionally consistent, although the prediction interval crossed no effect. Excluding the single systemic COX-2 inhibitor trial left seven local-anesthetic trials and 864 participants with a similar direction of effect (mean difference, -2.10; 95% confidence interval, -2.71 to -1.48; I-squared = 83.7%). Evidence was driven mainly by local anesthetic trials. Harms were sparsely and inconsistently reported. Certainty was low for pain reduction and very low for safety.

Conclusions: Low-certainty randomized evidence suggests that tested pain-control strategies, mainly local anesthetic interventions, may reduce average intra-procedural pain during blind office endometrial biopsy with inert comparators, but heterogeneity is substantial and the optimal intervention class remains uncertain. These findings support patient-centered discussion of pain-control options but do not establish any specific agent or intervention class as standard care.

Trial registration: Not applicable. This systematic review was not prospectively registered.

Keywords

Endometrial biopsy; endometrial sampling; procedural pain; lidocaine; local anesthesia; analgesia; systematic review; meta-analysis.

Introduction

Endometrial biopsy is a central office-based procedure for evaluating abnormal uterine bleeding and suspected endometrial pathology. It can usually be performed in the outpatient setting, provides direct histologic sampling of the endometrium, and is commonly used in the diagnostic workup for abnormal or postmenopausal bleeding [1-3]. Compared with operative sampling, blind outpatient biopsy avoids general anesthesia in many settings and can be integrated into routine gynecologic care. These practical advantages make the procedure valuable, but they also place the burden of pain control within the office encounter.

Pain during endometrial biopsy is clinically important. It can affect acceptability, patient satisfaction, and willingness to undergo future gynecologic procedures. Contemporary guidance on in-office uterine and cervical procedures emphasizes discussion of pain-management options and shared, patient-centered decision-making [4]. Observational evidence also shows that office gynecologic tract sampling produces measurable patient-rated pain, and that endometrial biopsy can be more painful than less invasive sampling approaches [5]. Evidence synthesis should therefore address not only whether the procedure can be performed efficiently, but whether pain scores can be lowered in a reproducible way.

The sources of biopsy pain are heterogeneous. Pain may arise from speculum placement, tenaculum application, cervical passage, uterine instrumentation, endometrial aspiration or sampling, and post-procedure cramping. Pain-control strategies target different parts of this pathway. Local anesthetics may be applied to the cervix, instilled into the uterine cavity, or injected as part of paracervical blockade. Systemic NSAIDs and COX-2 inhibitors may reduce prostaglandin-mediated cramping. Other adjuncts, including nonpharmacologic devices and cervical-preparation strategies, have also been tested [4,6]. This clinical diversity complicates synthesis, especially when procedure type, comparator credibility, and pain timepoint vary across trials.

Prior reviews of pain control for endometrial sampling, outpatient hysteroscopy, and related gynecologic procedures have had to balance broad coverage against clinical interpretability [6,7]. Charoenkwan and Nantasupha summarized a wider evidence base across procedures, interventions, comparators, and pain outcomes [6]. That breadth is useful for mapping the field, but it can make the pooled estimate difficult to apply to a single office procedure.

The present review therefore focused on a narrower clinical question: whether tested pain-control strategies lower pain scores during blind outpatient or office-based endometrial biopsy or sampling. We separated the headline pooled analysis from sensitivity and narrative evidence so that trials with non-extractable medians, binary-only outcomes, no-analgesic controls, co-intervention designs, nonpharmacologic adjuncts, cervical-preparation designs, or adjacent procedures remain visible without weakening the interpretation of the primary estimate.

The objective of this systematic review and meta-analysis was to evaluate randomized evidence for pain-control strategies during blind office endometrial biopsy, emphasizing procedural consistency, comparator credibility, extractable intra-procedural pain outcomes, risk of bias, certainty of evidence, safety reporting, and the implications of intervention heterogeneity.

Methods

Protocol and Registration

This systematic review and meta-analysis evaluates pain-control strategies for blind outpatient or office-based endometrial biopsy or endometrial sampling and follows PRISMA 2020 reporting principles [8]. The review was not prospectively registered. Prospective registration was no longer possible once data extraction and quantitative synthesis had already begun, so we chose transparent repository deposition rather than describing a retrospective record as prospective registration. The review question, eligibility criteria, outcome hierarchy, and analysis hierarchy were specified before quantitative synthesis, whereas some transparency-facing presentation decisions, such as expanded non-pooled evidence tabulation and repository packaging, were finalized during manuscript preparation. A protocol, analysis plan, extracted datasets, code, figures, and screening records were deposited in Zenodo before journal submission to improve transparency and reproducibility (Zenodo concept DOI: https://doi.org/10.5281/zenodo.20116870). Because the deposit occurred after completion of data extraction and quantitative synthesis, it should not be interpreted as prospective registration.

PICO Framework and Study Design

The eligibility framework was specified as follows. Population: women undergoing outpatient or office-based blind endometrial biopsy or endometrial sampling for any clinical indication. Intervention: peri-procedural analgesic, anesthetic, or pain-control strategies intended to reduce pain before or during biopsy or sampling. Comparator: placebo, normal saline, placebo gel, distilled water, or another credible inert control for the primary quantitative analysis; inert controls were defined as matched controls without intended analgesic pharmacologic activity. The term did not imply absence of possible mechanical, lubricating, or vehicle effects. No-anesthesia, usual-care, active-comparator, or co-intervention designs were retained only for broader qualitative or sensitivity assessment when relevant. Outcomes: the primary outcome was pain during biopsy or sampling on a 0-10 compatible scale; secondary or supplementary outcomes included high-pain thresholds, pain at other procedure-related timepoints, additional analgesia, satisfaction or willingness to repeat biopsy, procedure failure, and adverse events. Study design: randomized controlled trials were eligible for efficacy synthesis, whereas conference abstracts, registry entries, and secondary reports were used for citation mapping, boundary adjudication, or publication-bias assessment unless sufficient arm-level outcome data were available.

Eligibility Criteria

Randomized controlled trials were eligible for efficacy synthesis when they enrolled participants undergoing outpatient or office-based blind endometrial biopsy or endometrial sampling and reported pain, tolerability, additional analgesia, satisfaction, or adverse-event outcomes. No restrictions were placed on publication year, language, country, indication for biopsy, menopausal status, or biopsy device, provided that the target procedure was blind endometrial biopsy or sampling and the relevant pain outcome was separable. Conference abstracts, registry entries, and secondary reports were retained for citation mapping and publication-bias assessment when they described potentially eligible trials, but they entered quantitative synthesis only when sufficient arm-level outcome data were available.

Studies were excluded from the primary analysis when they centered on hysteroscopy-guided biopsy, operative hysteroscopy, dilation and curettage, abortion procedures, suction curettage or Vabra sampling without separable blind EMB pain data, or inseparable combined gynecologic procedures, because those settings differ in instrumentation, duration, uterine distension, cervical manipulation, and pain mechanisms. Device-comparison studies without a pain-control intervention, non-randomized studies, narrative reviews, guidelines, protocols without usable results, duplicate publications, and reports without usable pain or safety outcomes were excluded from review synthesis. Nonpharmacologic device adjuncts, cervical-preparation or procedure-facilitation interventions, active-comparator-only designs, no-treatment or usual-care comparator studies, and co-intervention designs were retained only as contextual or boundary evidence when clinically relevant and were not included in the primary inert-controlled mean-difference estimate.

For the primary continuous meta-analysis, comparisons had to meet all of the following additional criteria: the study had to be randomized; the procedure had to be blind endometrial biopsy or sampling rather than hysteroscopy-guided biopsy; the comparator had to be placebo, normal saline, placebo gel, or another credible inert control; the outcome had to be pain during biopsy or sampling, or the closest clearly intra-procedural biopsy pain timepoint; and the outcome had to be extractable as sample size, mean, and standard deviation on a 0-10 compatible pain scale, with higher scores indicating worse pain. Studies reporting medians and ranges only, interquartile ranges only, ordinal verbal rating scale distributions, binary high-pain thresholds, unresolved variance units, nonpharmacologic device adjuncts, or co-intervention designs not matching the primary estimand were not included in the primary estimate. When clinically relevant, such records were summarized narratively, tabulated as non-pooled evidence, or assigned to prespecified sensitivity analyses.

To avoid mixing clinically different comparisons, we organized the evidence into three tiers. Tier 1 was the prespecified primary meta-analysis. It included inert-controlled randomized comparisons with extractable intra-procedural means and standard deviations. Tier 2 comprised sensitivity analyses that tested analytic or clinical boundaries. These included balanced-background ibuprofen designs, in which both randomized groups received ibuprofen and the randomized contrast assessed the added topical or local intervention. Tier 2 also included the analysis excluding the single systemic-only COX-2 inhibitor trial. Tier 3 comprised narrative or tabulated contextual evidence. This tier included median-only, binary-only, ordinal-only, active-comparator, nonpharmacologic, cervical-preparation, co-intervention, suction/Vabra, registry-only, or mixed-procedure records. We applied this hierarchy before interpreting the pooled estimate.

Information Sources and Search Strategy

Database and register searches covered database inception to 2026-05-09, with Scopus and WHO ICTRP record capture completed on 2026-05-10. The PRISMA-counted search set included PubMed, CENTRAL, Embase, Web of Science Core Collection, Scopus, Wanfang, VIP, ClinicalTrials.gov, and WHO ICTRP. CNKI, SinoMed/CBM, and ChiCTR were checked as supplemental focused-search or registry-checking sources; because complete deduplicable exports or manual record tables were not part of the final PRISMA-counted dataset for those sources, they were not included in the main PRISMA identification denominator. No date, language, or publication-status filters were applied to the main bibliographic search strategies. Additional PubMed and ClinicalTrials.gov update checks were conducted on 2026-06-12 using the same core procedure, pain-control, and trial-design concepts; no additional trial eligible for the primary meta-analysis was identified.

The search strategies combined three concept blocks: the target procedure, the pain-control strategy, and randomized or controlled trial design. Database-specific syntax, full English and Chinese search strings, record counts, export or capture formats, and update checks are provided in the Supplementary Search Appendix and the repository package.

For Chinese sources, equivalent Chinese procedure, pain-control, and trial-design terms were used with platform-appropriate fields. Wanfang and VIP contributed deduplicable record captures to the PRISMA-counted dataset, whereas CNKI and SinoMed/CBM were retained as supplemental focused-search sources because comprehensive auditable exports were not obtained.

ClinicalTrials.gov and WHO ICTRP were searched with simplified procedure and pain-control terms to maximize sensitivity. ChiCTR was checked with manual Chinese and English registry terms and retained as supplemental registry audit because no locked exportable or manually deduplicable record table was created.

Study Selection and Data Extraction

Search results were deduplicated using PMID, DOI, NCT number, trial title, normalized title strings, and other available identifiers. Records selected for further review were collapsed into study-family records so that duplicate database hits, registry-publication links, secondary reports, and title variants would not be double counted. Registry-publication links were tracked separately for duplicate handling and publication-bias assessment.

Eligibility was assessed at title/abstract and full-text stages, with uncertain records retained for full-text or boundary review. Screening and eligibility-critical decisions were checked by rotating reviewer pairs drawn from three reviewers: Bin Yue, Cui Zhao, and Xiaogang Hao. At least two reviewers verified records retained as include, unclear, full-text, or boundary decisions. Full-text exclusions, primary-analysis inclusion, effect-size extraction, and RoB 2 judgments were also checked by two reviewers. Disagreements were resolved by discussion, with senior clinical and methodological input from Zhirong Dai when needed. At full-text review, studies were assessed for procedure setting, blind versus hysteroscopy-guided biopsy, randomized or controlled design, intervention relevance, comparator type, extractable outcome data, duplicate publication status, and whether EMB-related pain was separable from other procedures.

For the paired human verification subset, Cohen's kappa was calculated when two reviewer decisions were available. Because the workflow also used machine prescreening, staged human verification, and consensus adjudication, this kappa estimate describes that paired verification subset and should not be interpreted as full independent duplicate screening of all 2,013 title/abstract records.

Data were extracted with structured study-, arm-, outcome-, and analysis-level forms. One reviewer extracted each record and a second reviewer checked eligibility classification, arm denominators, outcome timepoint, mean, standard deviation, direction of effect, intervention/comparator label, and analysis assignment. Extracted study-level fields included authorship, publication year, setting, country, design, sample size, eligibility criteria, biopsy device, procedural context, intervention details, comparator details, randomization and blinding information, registration or protocol availability, and funding or conflict-of-interest information when reported. Arm-level extraction captured the active intervention, dose or concentration, timing, route, co-interventions, comparator content, and number randomized or analyzed. Each relevant record was classified as included in the primary analysis, summarized narratively, assigned to sensitivity analysis, or excluded from the primary estimand.

For continuous pain outcomes, extraction prioritized pain during biopsy or sampling. When several pain timepoints were available, the closest clearly intra-procedural biopsy or sampling pain timepoint was selected. Extracted fields included the pain scale, timepoint, sample size, mean, standard deviation, and direction of scoring. For dichotomous outcomes, extraction captured event thresholds, event counts, denominators, timing, and whether thresholds represented severe, unacceptable, or otherwise high pain. Safety extraction captured serious complications, procedure-related complications, vasovagal reactions, hypotension, dizziness, adverse effects, procedure failure, and any author statement indicating no adverse events or no serious complications.

When a trial contributed more than one eligible active arm against a single inert control, active arms were combined before meta-analysis so that the shared control group was counted only once. Combined means were calculated as sample-size weighted means. Combined standard deviations were calculated from arm-level sample size, mean, and standard deviation using the Cochrane Handbook formula for combining groups. The pooled active arm was then compared with the original control arm for that study. This approach was used only when the active arms belonged to the same prespecified pain-control family for the primary estimand and when arm-level mean, standard deviation, and analyzed sample size were available.

Outcomes

The primary outcome was pain during endometrial biopsy or endometrial sampling, measured on a 0-10 compatible visual analogue, numerical rating, or equivalent pain scale, with higher scores indicating worse pain. The primary effect direction was active intervention minus comparator; negative mean differences therefore favored the intervention.

Secondary and supplementary outcomes included binary high-pain outcomes, pain at other procedure-related timepoints, need for additional analgesia, participant satisfaction or willingness to undergo repeat biopsy, procedure failure, and adverse events or complications. Safety outcomes were prespecified for narrative synthesis unless trial definitions and surveillance methods were sufficiently consistent for meaningful quantitative pooling.

Risk of Bias and Certainty of Evidence

Risk of bias for randomized trials was assessed using the Cochrane RoB 2 framework at the outcome level, with primary emphasis on pain during biopsy or sampling [9]. RoB 2 judgments for the primary-analysis trials were made or checked by two reviewers and finalized by consensus. Particular attention was given to allocation concealment, participant and operator blinding, credibility of placebo or sham controls, post-randomization exclusions, completeness of pain-score reporting, and whether multiple pain timepoints created selective-reporting concerns. Study-level domain judgments and their contribution to the outcome-level GRADE assessment are provided in Supplementary Table S3.

Certainty of evidence was assessed using the GRADE approach for outcomes prioritized for the Summary of Findings [10]. Randomized trials started as high-certainty evidence and were considered for downgrading due to risk of bias, inconsistency, indirectness, imprecision, and publication bias. GRADE certainty was assessed at the outcome level rather than as a separate certainty rating for each individual study. For pain during biopsy, downgrading was driven by RoB 2 some-concern judgments in five primary trials and substantial heterogeneity. For safety, downgrading was driven by sparse event data, inconsistent adverse-event definitions and surveillance windows, incomplete arm-level reporting, and small trial sizes; therefore safety certainty was rated very low. To improve transparency, we added a supplementary study-level table linking each primary-analysis trial to its RoB 2 judgment and contribution to the GRADE rationale.

Effect Measures and Synthesis Methods

For continuous pain outcomes on a 0-10 compatible scale, the effect measure was the mean difference. Scores reported on a 0-100 mm scale may be linearly expressed on a 0-10 scale when otherwise eligible, but the primary analysis did not use distributional conversions from medians, ranges, interquartile ranges, or ordinal categories. Standard errors for study-specific mean differences were calculated from extracted arm-level sample sizes and standard deviations when not directly available. For binary high-pain and adverse-event outcomes, the planned effect measure was the risk ratio with 95% confidence intervals.

The primary synthesis used a random-effects meta-analysis of mean differences for trials with placebo, saline, or placebo-gel controls and extractable intra-procedural pain scores on a 0-10 compatible scale. The conventional headline analysis used a DerSimonian-Laird random-effects model based on study-level mean differences and standard errors. Statistical heterogeneity was assessed using the Q statistic, tau-squared, and I-squared. Because the number of studies was small and heterogeneity was substantial, robustness was checked using restricted maximum likelihood random-effects estimation, Hartung-Knapp-Sidik-Jonkman confidence intervals, and 95% prediction intervals, and these checks were used to judge the uncertainty around the average effect. Heterogeneity was expected because the review question covers different pain-control mechanisms within a shared office biopsy setting.

Sensitivity analyses tested whether the direction and magnitude of the primary estimate were robust to analytic boundary decisions. Leave-one-out analysis was used to assess whether any single study disproportionately influenced the primary estimate. Additional sensitivity analyses included exclusion of high-risk-of-bias studies, exclusion of explicitly non-blinded or likely unmasked no-analgesic controls, restriction to placebo or saline controls, and an intervention-boundary analysis excluding the single systemic-only COX-2 inhibitor trial. A separate balanced-background-ibuprofen sensitivity analysis added trials in which both randomized groups received oral ibuprofen before biopsy; this analysis was used to examine whether the added value of topical local anesthetic remained directionally consistent on an NSAID background and was not part of the primary estimand. To explore clinical heterogeneity, we also generated an exploratory forest plot stratified by intervention mechanism: intrauterine local anesthetic, cervical topical anesthetic, mixed local anesthetic, and systemic COX-2 inhibition. Random-effects class summaries were shown only when at least two trials were available. Because class-specific groups were small, these estimates were descriptive; no formal test of subgroup differences or meta-regression was performed.

The primary estimand was the random-effects mean difference for inert-controlled randomized comparisons reporting extractable intra-procedural pain scores on a 0-10 compatible scale. Eligible inert comparators included placebo, saline, placebo gel, distilled water, or another inactive control. Sensitivity analyses examined the stability of this average estimate under alternative analytic boundaries. The balanced-background-ibuprofen analysis included trials in which both groups received ibuprofen and the randomized contrast isolated the added value of topical lidocaine gel. The intervention-boundary analysis excluded the single systemic-only COX-2 inhibitor trial. Trials outside the primary estimand or without extractable means and standard deviations were summarized narratively. This included trials reporting only ordinal or binary pain outcomes. Analyses were conducted in R version 4.6.0. Random-effects meta-analysis, leave-one-out analysis, REML estimation, Hartung-Knapp-Sidik-Jonkman intervals, and prediction intervals used metafor 5.0-1 and meta 8.5-0. Data handling used readr, dplyr, stringr, and related tidyverse packages as recorded in the session information. The updated PRISMA flow diagram, balanced-background-ibuprofen sensitivity plot, COX-2 exclusion sensitivity plot, and exploratory mechanism-stratified forest plot were generated from locked CSV files using scripted graphics. Earlier primary forest, leave-one-out, and RoB 2 figures were retained from the reproducibility package, with package information provided in the repository.

Publication Bias and Reporting Bias

Because the primary analysis included fewer than 10 trials, funnel plots and Egger's regression test were not performed. These assessments have limited power and unstable interpretation when so few studies are available. Publication bias therefore remained unresolved rather than excluded. Reporting bias was assessed qualitatively by comparing trial registries, posted results, mapped publications, and source-data availability (Supplementary Table S4).

Registry searches identified completed or status-unknown records without posted results or clearly mapped publications. The most directly relevant example was NCT02465320, a completed phase 2 randomized, double-blind, placebo-controlled study of COL-1077 10% lidocaine bioadhesive gel versus placebo in women undergoing transvaginal Pipelle-directed endometrial biopsy; the registry lists 187 enrolled participants and no posted results [11]. This uncertainty was considered when rating publication bias in GRADE.

Results

Search Results and Study Selection

The PRISMA-counted search dataset identified 2,632 records from databases and registers, comprising PubMed, CENTRAL, Embase, Web of Science, Scopus, Wanfang, VIP, ClinicalTrials.gov, and WHO ICTRP. After removal of 619 duplicate records, 2,013 records remained for title and abstract screening; 1,805 records were excluded at title/abstract screening because they did not address a relevant pain-control intervention, evaluated hysteroscopy or other non-target procedures, were device-only or diagnostic-accuracy records, were reviews or guidelines, involved non-human or laboratory settings, or lacked usable pain/tolerability information. Of 208 reports or records sought for retrieval or eligibility clarification, 10 were not retrieved or had no public extractable results and 198 were assessed for eligibility. Of these, 180 were excluded or not included in the result-bearing synthesis. The leading full-text or boundary categories included hysteroscopy or mixed procedures without separable EMB pain data, no public extractable results, nonpharmacologic or procedural comfort adjuncts outside the prespecified analgesia/anesthesia estimand, cervical-preparation or procedure-facilitation interventions, active-comparator or non-inert comparator designs, co-intervention designs, and non-extractable median, ordinal, or binary outcome reporting (Fig. 1; Supplementary Table S5). Eighteen study families or result-bearing trials were included in the review synthesis: eight primary-analysis RCTs, one balanced-background sensitivity RCT, and nine narrative, non-pooled, or boundary RCTs.

In the paired human verification subset with two reviewer decisions available (n = 256), observed agreement was 229/256 (89.5%) and Cohen's kappa was 0.830. This value should be interpreted as agreement within the paired verification subset rather than as duplicate independent screening of the full machine-prescreened record set.

The quantitative efficacy synthesis was narrower than the full set of records undergoing further review. The primary meta-analysis was restricted to randomized controlled trials evaluating pain-control interventions during blind office or outpatient endometrial biopsy or endometrial sampling, using placebo, saline, placebo gel, or another inert comparator, and reporting extractable mean and standard deviation data for pain during biopsy or the closest clearly intra-procedural biopsy pain timepoint on a 0-10 compatible scale. Eight randomized trials, contributing 944 participants, met these primary-analysis criteria.

Study Characteristics

The eight primary trials evaluated pain-control strategies within blind outpatient or office-based endometrial biopsy or sampling settings (Table 1). The interventions were clinically heterogeneous but shared the target procedure and outcome framework. Two trials evaluated intrauterine local anesthetic strategies, one evaluated a mixed local anesthetic strategy, four evaluated cervical topical local anesthetic spray or gel, and one evaluated an oral COX-2 inhibitor. Comparators included saline control, distilled water placebo, normal saline, placebo tablet, placebo spray, and placebo gel.

The primary studies were Kosus 2014, Olad-Saheb-Madarek 2013, Benchahong 2018, Tanprasertkul 2008, Aksoy 2015, Karaca 2016, Korsuwan 2018, and Likkasittipan 2021 [12-19]. Kosus 2014 compared intrauterine levobupivacaine plus intrauterine lidocaine with saline control in outpatient blind endometrial biopsy [12]. Olad-Saheb-Madarek 2013 compared active local anesthetic strategies, including intrauterine lidocaine, cervical lidocaine spray, or combined active local anesthetic arms, with distilled water placebo in outpatient Pipelle endometrial biopsy [13]. Benchahong 2018 compared intrauterine lidocaine with normal saline during outpatient blind endometrial biopsy with a Novak device [14]. Tanprasertkul 2008 compared etoricoxib 120 mg with placebo tablet [15]. Aksoy 2015 compared 10% lidocaine spray with placebo spray in office-based endometrial sampling, using the pain-analysis denominator of 60 versus 59 participants after one failed placebo-group procedure [16]. Karaca 2016 compared 2% cervical lidocaine gel with placebo gel during Pipelle endometrial sampling [17]. Korsuwan 2018 compared 10% lidocaine spray with placebo spray during Wallach Endocell office-based endometrial biopsy; one text passage differed from the abstract/table denominator, so the table/abstract denominator was used with an extraction note [18]. Likkasittipan 2021 compared 2% lidocaine gel with placebo gel during outpatient endometrial sampling with Wallach Endocell [19]. Likkasittipan 2021 explicitly reported pain during endometrial aspiration. Korsuwan 2018 labelled the selected timepoint as during endometrial biopsy in its table and figure but described the corresponding score in the text as pain during the procedure. We retained the Korsuwan value as the closest clearly intra-procedural biopsy timepoint and flagged this source-wording ambiguity in Table 1.

Risk of Bias

For the primary pain outcome, three of the eight primary trials were judged at low risk of bias and five had some concerns under RoB 2. Benchahong 2018, Tanprasertkul 2008, and Likkasittipan 2021 were judged low risk. Kosus 2014, Olad-Saheb-Madarek 2013, Aksoy 2015, Karaca 2016, and Korsuwan 2018 had some concerns. The main concerns were incomplete reporting of allocation concealment, incomplete information on trial registration or outcome selection, possible operator awareness of treatment assignment, and post-randomization exclusions or denominator issues. No trial judged at high risk of bias was included in the primary analysis.

Primary Pain Outcome

Compared with inert controls, pain-control interventions were associated with lower average procedural pain scores during blind office or outpatient endometrial biopsy (Fig. 2; Table 2). Risk-of-bias judgments for the primary-analysis trials are shown in Fig. 3. In the primary random-effects meta-analysis of eight RCTs and 944 participants, the pooled mean difference was -1.87 points on a 0-10 pain scale (95% CI, -2.52 to -1.22). Negative values favored the intervention. The confidence interval excluded no effect and was compatible with a potentially clinically relevant average reduction, but certainty was low and the estimate should not be interpreted as the effect of any single intervention class.

Statistical heterogeneity was substantial ($I^2 = 86.7\%$). This heterogeneity is consistent with clinical diversity in intervention mechanism, including intrauterine local anesthetic instillation, cervical topical anesthetic spray or gel, a mixed local anesthetic approach, and systemic COX-2 inhibition. It also reflects variation in biopsy devices, comparator formats, and intra-procedural pain measurement. The pooled estimate should therefore be interpreted as an average across clinically different intervention classes rather than as the effect of one specific strategy.

Sensitivity Analyses

The result was directionally stable in leave-one-out analyses. Removing one study at a time yielded pooled mean differences ranging from -2.10 to -1.61, and all recalculated estimates continued to favor active intervention. Heterogeneity remained present after each omission, with I^2 ranging from 74.4% to 88.6%, indicating that inconsistency was not driven by a single outlying trial.

A conservative sensitivity analysis retained the earlier six-trial dataset before addition of the two newly identified Thai placebo-controlled trials. This analysis included six RCTs and 836 participants and produced a similar direction of effect (MD -1.61; 95% CI, -2.17 to -1.06; I^2 = 78.7%).

A supplementary sensitivity analysis added Tueman 2026, a published double-blind placebo-gel trial in which both groups received oral ibuprofen before biopsy [20]. This balanced-background-ibuprofen analysis included nine RCTs and 1,030 participants and produced a similar estimate (MD -1.89; 95% CI, -2.48 to -1.30; I^2 = 84.8%). Because both groups received background ibuprofen, this sensitivity analysis was interpreted as supportive evidence for the added value of topical lidocaine gel on an NSAID background rather than as part of the primary estimand; its placement reflects an estimand boundary decision, not an evolving dataset.

An intervention-boundary sensitivity analysis excluded Tanprasertkul 2008, the only systemic-only COX-2 inhibitor trial in the primary analysis. The remaining seven local-anesthetic trials included 864 participants and continued to favor active intervention (MD -2.10 points; 95% CI, -2.71 to -1.48; I-squared = 83.7%). This analysis did not remove heterogeneity, but it showed that the headline direction was not dependent on the near-null systemic COX-2 trial and that the primary estimate was driven mainly by local anesthetic evidence.

Because class-specific groups were small, mechanism-stratified estimates were treated as exploratory and no formal test of subgroup differences was performed (Supplementary Figure S5). The two intrauterine local-anesthetic trials (340 participants) yielded an exploratory pooled MD of -1.42 (95% CI, -1.86 to -0.98; I-squared = 0.0%), whereas the four cervical topical-anesthetic trials (364 participants) yielded an exploratory pooled MD of -2.21 (95% CI, -3.15 to -1.26; I-squared = 87.7%). The mixed local-anesthetic and systemic COX-2 categories each contained one trial and were not pooled within class. These descriptive results indicate that intervention mechanism explains some between-class variation but does not account for the substantial inconsistency among cervical topical trials; class-specific estimates should therefore not be interpreted as comparative treatment rankings.

Robust random-effects checks supported the direction of the main result but emphasized the width of between-study variation. Using REML, the primary estimate was MD -1.87 (95% CI, -2.58 to -1.15). With Hartung-Knapp-Sidik-Jonkman adjustment, the confidence interval widened but still excluded no effect (MD -1.87; 95% CI, -2.73 to -1.00). The 95% prediction interval was wide and crossed no effect (-4.37 to 0.64), indicating that a future comparable trial could plausibly show a smaller or null effect because intervention class and procedure details vary. These robustness checks therefore support cautious inference: the average effect remained similar across models, but between-study variation means that future study-level effects may be smaller or null. In the balanced-background-ibuprofen sensitivity analysis, REML and Hartung-Knapp results were also directionally consistent (HKSJ 95% CI, -2.63 to -1.14), while the prediction interval again crossed no effect (-4.12 to 0.35).

Non-Pooled Relevant Trials

Several clinically relevant trials were not included in the primary meta-analysis because they did not meet the extractability or estimand requirements for the primary continuous pain synthesis (Table 3). Sripha 2018 was a relevant double-blind placebo-controlled office endometrial biopsy trial of 10% lidocaine spray versus saline placebo, but the available continuous pain outcome was reported as median and range rather than mean and standard deviation [21]. We therefore summarized it narratively and did not convert it for the primary analysis. The reported median VAS during cannula insertion was 5.0 versus 5.5, with p = 0.78.

Luangtangvarodom 2018 was a double-blind placebo-controlled trial of 10% lidocaine spray versus normal-saline spray during outpatient Novak endometrial sampling [22]. It was not eligible for the continuous mean/SD primary pool because the relevant extractable result was a severe-pain threshold rather than arm-level mean and standard deviation data. The reported severe pain rates during tissue aspiration were 42/70 with lidocaine spray versus 55/70 with saline placebo. Piyawetchakarn 2019 was a randomized three-arm trial of lidocaine spray, placebo spray, and no intervention during Pipelle endometrial aspiration biopsy [23]. It was retained narratively because biopsy pain was reported as median and range: 5.1 (0.1-10.0) with lidocaine spray and 5.8 (0.0-10.0) with placebo spray; no median-to-mean conversion was used.

Chayuttawanichakul 2022 was also a relevant double-blind placebo-controlled lidocaine-gel trial in women undergoing endometrial sampling, but the pain outcomes were reported as medians across procedural timepoints rather than arm-level means and standard deviations [24]. Hui 2006 was considered clinically relevant but non-pooled [25]. The available source confirmed an intrauterine lignocaine versus saline comparison during endometrial sampling or suction curettage, but the currently available data come from a secondary abstract or commentary source and provide median VAS values rather than extractable mean and standard deviation data. Hui 2006 was therefore summarized narratively only. The available median VAS values were 2.3 versus 4.2, with $p < 0.01$. Siddle 1983 was retained as a historical double-blind naproxen versus placebo Vabra suction-curettage record, but the available abstract did not provide arm-level mean and standard deviation data for pain during the procedure [26].

We also cross-checked known trials from prior reviews that readers may expect to see considered. Trolice 2000 was a randomized double-masked Pipelle endometrial biopsy trial of intrauterine lidocaine versus saline, but the indexed report summarized pain using median and range values on a 20-cm VAS and was not pooled in the mean/SD primary analysis [27]. Kozman 2001 was a randomized double-blind placebo-controlled Vabra sampling trial of 2% lignocaine gel versus inert gel, but the main extractable result was a binary high-pain threshold rather than a continuous mean/SD outcome compatible with the primary estimand [28]. Kaya 2015 tested intrauterine lidocaine with or without rectal diclofenac against placebo in office endometrial biopsy, but it was retained as boundary evidence because the active co-intervention structure and extractability did not fit the prespecified placebo-controlled mean/SD primary pool [29]. These records are listed in the supplementary prior-review comparison and non-pooled evidence tables.

Other relevant records were not included in the primary estimate when they involved hysteroscopy-only or mixed hysteroscopy and biopsy procedures without separable blind EMB pain data, active-comparator-only designs, no-analgesic controls, ordinal-only outcomes, registry-only records without usable results, nonpharmacologic adjuncts outside the primary estimand, or unresolved variance reporting. This approach preserved procedural consistency, comparator credibility, and a common continuous pain-score outcome for the primary synthesis.

Safety and Certainty of Evidence

Safety reporting was sparse and inconsistent across trials (Supplementary Table S2). All eight primary-analysis trials either provided an explicit safety statement or reported selected adverse-event or complication items, but definitions, follow-up windows, and arm-level event counts differed. Examples included hypotension requiring positioning, dizziness, vasovagal reactions, procedure failure related to cervical stenosis, minimal cervical bleeding, and statements that procedures were completed without complications or without serious lidocaine-related reactions. No clear serious-harm signal was identified in the randomized evidence reviewed for the prespecified hierarchy, but the available data are insufficient to establish precise comparative safety.

The certainty of evidence for the primary pain outcome was rated as low. Downgrading was driven by risk of bias and inconsistency. Five of the eight primary trials had some concerns in RoB 2 assessment, and the pooled result had substantial heterogeneity. Indirectness was not judged serious within the prespecified scope of blind office/outpatient EMB or closely comparable blind sampling. Imprecision was not judged serious because the confidence interval remained entirely on the benefit side and compatible with clinically meaningful pain reduction. Publication bias remains unresolved because the number of pooled trials is small and registry-publication mapping identified unpublished or unmapped trial signals.

Although publication bias remains unresolved and one registry record without posted results (NCT02465320) was identified, we did not apply an additional GRADE downgrade for this domain. GRADE downgrading is warranted when publication bias is strongly suspected. In this review, the effect direction of the unavailable registry result was unknown, and the small number of pooled trials precluded a reliable assessment of small-study effects. The available information was therefore insufficient to establish strongly suspected publication bias. This was an uncertain domain judgment, not evidence that publication bias was absent, and the unresolved risk was retained explicitly in the Summary of Findings and limitations.

Discussion

Principal Findings

This review suggests that pain during blind outpatient or office-based endometrial biopsy may be modifiable, but the evidence should be interpreted cautiously. In placebo-, saline-, or placebo-gel-controlled trials, active pain-control strategies were associated with an average reduction of approximately 1.9 points on a 0-10 intra-procedural pain scale. The finding was directionally stable in leave-one-out analysis and remained similar in the conservative six-trial and balanced-background-ibuprofen sensitivity analyses. However, the certainty of evidence was low, heterogeneity was substantial, and the prediction interval crossed no effect.

The observed average difference is compatible with a clinically relevant reduction for many patients, but clinical importance should not be reduced to a single numeric threshold. In acute pain studies using visual analogue scales, minimum clinically important differences have often been reported around 9 to 13 mm on a 100-mm scale, roughly 0.9 to 1.3 points on a 0-10 scale [30,31]. The pooled mean difference of 1.87 points exceeds these commonly cited thresholds, yet IMMPACT guidance emphasizes that interpretation should also consider baseline pain, patient burden, available alternatives, adverse events, and the risk-benefit profile [32]. In this review, the clinical meaning of the pooled estimate is limited by intervention heterogeneity, low certainty, and incomplete safety reporting.

Procedural Consistency and Intervention Heterogeneity

The most important methodological decision in this review was to prioritize procedural consistency. Blind office endometrial biopsy produces a different pain experience from hysteroscopy-guided biopsy, operative hysteroscopy, dilation and curettage, abortion procedures, or combined intrauterine interventions. Those procedures differ in instrumentation, cervical manipulation, uterine distension, duration, operator control, and patient expectations. Pooling them would increase study numbers at the cost of interpretability.

For that reason, the primary analysis was framed around blind outpatient or office-based endometrial biopsy rather than all endometrial procedures. This narrower framing makes the result more clinically useful. The findings apply most directly to blind office biopsy and should not be generalized uncritically to other gynecologic procedures.

The high I^2 value is an important limitation, but it does not negate the consistent direction of effect across the primary trials. REML and Hartung-Knapp checks gave similar average effects, while the prediction interval crossed no effect, which is the expected pattern when an average benefit is present but study-level effects vary widely. The included interventions are mechanistically diverse, which largely accounts for the high residual heterogeneity even when restricting to local anesthetics. Specifically, intrauterine local anesthetics target nociception arising from endometrial sampling or uterine cavity distension, whereas cervical sprays and gels primarily mitigate pain from tenaculum application and cervical instrument passage. Systemic NSAIDs or COX-2 inhibitors may influence prostaglandin-mediated cramping rather than immediate nociception. Mixed local anesthetic approaches combine more than one potential target.

A further methodological consideration regarding comparator credibility is the physical property of the inert controls. While we categorized normal saline, distilled water, and placebo gels collectively as pharmacologically inert, the gel vehicle itself may confer a physical lubricating effect. This mechanical lubrication could theoretically reduce friction during cervical instrument passage and lower pain independently of the active drug. In gel-controlled trials, the active and placebo groups shared this vehicle, so any lubricating effect would be expected to attenuate the pharmacologic contrast rather than exaggerate it. Across the meta-analysis, however, differences between gel, spray, saline, water, and tablet comparators introduce additional control-arm heterogeneity. Placebo gels should therefore be understood as pharmacologically inert, but not necessarily mechanically inert.

This diversity means the pooled estimate should not be read as a recommendation for a single best intervention. It is better interpreted as an average effect across a family of pain-control strategies tested in a shared clinical procedure. Descriptively, the clearest pool of evidence in the current primary analysis comes from cervical topical anesthetic spray or gel studies, but those trials still vary in product, dose, device, and denominator handling. Intrauterine anesthetic studies are fewer and clinically plausible, while the systemic NSAID or COX-2 evidence is represented by a single primary-analysis trial. Future syntheses are most informative when they stratify interventions by mechanism, including intrauterine local anesthetic, cervical topical anesthetic, paracervical block, systemic NSAID, cervical preparation, and nonpharmacologic adjuncts.

The oral COX-2 inhibitor trial illustrates why the pooled estimate should not be read as proof that every intervention class is effective. Tanprasertkul 2008 compared etoricoxib 120 mg with placebo in 40 participants per arm and had a near-null during-biopsy estimate (MD -0.17 points; 95% CI, -1.04 to 0.70). This result may reflect the intervention mechanism, timing, sample size, or limited comparability with local anesthetic trials. It should not by itself rule out systemic analgesia, but it supports treating intervention class as a key source of heterogeneity rather than making a single class-level recommendation.

Interpretation of Non-pooled Evidence

Several eligible or potentially relevant studies were not included in the primary analysis because they did not meet the prespecified extractability or estimand criteria. Sripha 2018, Piyawetchakarn 2019, and Chayuttawanichakul 2022 were placebo-controlled office endometrial sampling trials, but the available pain outcomes were reported as medians rather than arm-level means and standard deviations. Luangtangvarodom 2018 provided binary severe-pain data rather than a continuous mean/SD endpoint. Hui 2006 was clinically relevant, but the available source was a secondary abstract or commentary source with median VAS data rather than the original arm-level mean and standard deviation data; it was therefore not used to support the pooled estimate. Siddle 1983 and Kozman 2001 involved Vabra or suction-curettage sampling and were retained as boundary/contextual evidence rather than primary blind EMB evidence.

Other boundary decisions followed the same principle. Trials reporting only medians and ranges were not converted into means and standard deviations for the main result because such conversion would require distributional assumptions. Ordinal verbal rating scale studies were not mixed with continuous VAS or NPRS outcomes. Somchit 2015 was not included because the reported variance values for pain during biopsy are implausibly small as standard deviations and are more consistent with standard errors or a reporting problem. TENS was treated as a nonpharmacologic device adjunct rather than a pharmacologic or local anesthetic intervention. Guney 2007 was held as a co-intervention study because all participants received buccal misoprostol, and subgroup denominators were not clearly extractable from the available report.

These decisions reduce apparent inclusiveness but improve interpretability. The primary analysis prioritized a clinically interpretable estimate for a specific procedure and outcome over a broader but less coherent pooled dataset.

Safety and Clinical Application

Available safety data did not reveal a clear signal of serious harm, but the certainty of this conclusion is very low. Serious complications were not identified in trials that explicitly described them, and reported non-serious events were uncommon or variably defined. However, most trials were small, event definitions differed, follow-up windows were short or unclear, and adverse-event collection was not consistently detailed. The absence of a clear serious-harm signal should therefore be interpreted as reassuring but not definitive, and it should not be used as evidence that all pain-control strategies have equivalent safety.

Clinically, the findings support discussing pain-control options with patients undergoing office endometrial biopsy, while acknowledging that the current evidence does not establish one universally superior strategy. In settings where office endometrial biopsy is performed without routine local anesthesia or a standardized analgesic protocol, the practical implication is not that one method should become standard care immediately, but that pain-control options should be explicitly considered as part of shared procedural planning. The choice of approach should account for onset time, visit logistics, patient preference, contraindications, equipment needs, background analgesic use, and local familiarity. The pooled result is best used to support patient-centered procedural planning, not as a directive for a single intervention.

Limitations

This review has several limitations. First, clinical and intervention heterogeneity was substantial. The primary pool combined intrauterine local anesthetic strategies, cervical topical spray or gel, a mixed local anesthetic approach, and one systemic COX-2 inhibitor trial. These interventions target different pain mechanisms and should not be interpreted as interchangeable.

Second, methodological limitations in the included trials reduced certainty. Five of the eight primary studies had some concerns in RoB 2 assessment, most often because of incomplete allocation-concealment information, possible operator awareness of treatment assignment, missingness or denominator issues, or limited auditability of outcome selection.

Third, several clinically relevant studies could not be included in the continuous primary estimate because they reported medians, ranges, ordinal outcomes, binary high-pain thresholds, ambiguous variance values, or estimands outside the prespecified primary comparison. This restriction may selectively emphasize trials with extractable continuous mean/SD data. These studies remain important contextual evidence, but mixing them into the main mean-difference pool would have required additional assumptions.

Fourth, safety evidence was sparse and inconsistently reported. Adverse events were not defined or collected consistently, and serious events were rare or absent in small trials. The certainty of the safety evidence was therefore very low, and the available data do not permit definitive conclusions about the comparative harms of these interventions.

Fifth, publication bias remains unresolved. Because the primary meta-analysis included fewer than 10 trials, funnel plots and Egger's regression test were not performed. Registry-publication mapping also identified unresolved signals, so publication bias cannot be excluded.

Sixth, the review was not prospectively registered. Although the protocol, analysis plan, extracted datasets, code, figures, and screening records were deposited in Zenodo before journal submission, this deposit occurred after data extraction and quantitative synthesis. Readers should therefore distinguish repository transparency from prospective registration.

Seventh, geographic generalizability is limited. Most primary-analysis trials were conducted in Thailand or Turkey, with additional evidence from Iran and other settings; therefore the estimates may not fully represent pain expectations, biopsy practices, background analgesic use, or office workflow in all health-care systems. Chinese databases were searched with Chinese procedure and pain-control terms, but Wanfang and VIP were the only Chinese sources included in the deduplicable PRISMA-counted dataset. CNKI, SinoMed/CBM, and ChiCTR were retained as supplemental checks because complete auditable exports were not obtained, which remains a limitation of the search documentation.

Eighth, the wording of the intra-procedural pain timepoint was not fully uniform across the primary literature. Likkasittipan 2021 explicitly reported pain during endometrial aspiration, whereas Korsuwan 2018 labelled the selected timepoint as during endometrial biopsy in its table and figure but described it as pain during the procedure in the accompanying text. We retained the Korsuwan value under the prespecified rule selecting the closest clearly intra-procedural biopsy timepoint. This limited source-wording ambiguity may contribute to clinical heterogeneity and prevents treating every trial's pain timepoint as perfectly identical.

Together, these limitations mean that the pooled estimate should be treated as a cautious average across a heterogeneous family of pain-control strategies. Future randomized trials should use prospective registration, clearly matched comparators, prespecified intra-procedural pain endpoints, standardized adverse-event reporting, transparent allocation concealment, and complete reporting of means and standard deviations or re-analysable raw data.

Conclusions

Low-certainty randomized evidence suggests that tested pain-control strategies, mainly local anesthetic interventions, may lower average intra-procedural pain scores during blind outpatient or office-based endometrial biopsy compared with inert controls. However, intervention heterogeneity was substantial, the prediction interval crossed no effect, safety certainty was very low, and available data do not identify an optimal intervention class or establish any strategy as standard care. Future trials should use matched comparators, standardized pain endpoints, transparent allocation concealment, prospective registration, and complete adverse-event reporting.

List of abbreviations

AE, adverse event; CI, confidence interval; EMB, endometrial biopsy; GRADE, Grading of Recommendations Assessment, Development and Evaluation; MD, mean difference; NSAID, non-steroidal anti-inflammatory drug; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; RCT, randomized controlled trial; RoB 2, revised Cochrane risk-of-bias tool for randomized trials; SAE, serious adverse event; SoF, Summary of Findings; VAS, visual analogue scale.

Declarations

Ethics approval and consent to participate

Not applicable. This study is a systematic review and meta-analysis of previously published or registered studies.

Consent for publication

Not applicable.

Availability of data and materials

Data, code, search documentation, screening records, risk-of-bias/GRADE materials, figures, and reproducibility files are available at Zenodo [33]: https://doi.org/10.5281/zenodo.20116870.

Additional files

Additional file 1. Reproducibility package (.zip): protocol and analysis plan, search strategies and log, PRISMA counts, family-level full-text/boundary disposition table, extracted datasets, risk-of-bias and GRADE materials, analysis code, tables, and generated figures.

Additional file 2. PRISMA 2020 checklist (.docx): completed checklist with manuscript section references.

Additional file 3. Supplementary materials: Supplementary Search Appendix, Supplementary Table S2 safety reporting, Supplementary Table S3 study-level RoB 2 and GRADE contribution, Supplementary Table S4 reporting-bias assessment, Supplementary Table S5 documented exclusion categories, Supplementary Table S6 sensitivity analyses including exclusion of the systemic-only COX-2 inhibitor trial, Supplementary Figure S1 leave-one-out sensitivity, Supplementary Figure S2 balanced-background-ibuprofen sensitivity, Supplementary Figure S4 COX-2 exclusion sensitivity plot, and Supplementary Figure S5 exploratory mechanism-stratified forest plot.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

BY, CZ, and XH contributed equally to the study conception, literature screening, full-text eligibility assessment, data checking, RoB 2 checks, data interpretation, and manuscript writing. Screening and data-checking tasks were performed in rotating reviewer pairs. HK, XY, MZ, XJ, QW, and XZ contributed to literature screening, data checking, clinical interpretation, and critical revision of the manuscript. ZD supervised the study, helped resolve methodological and clinical questions, and is the corresponding author. All authors read and approved the final manuscript.

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Figure Legends

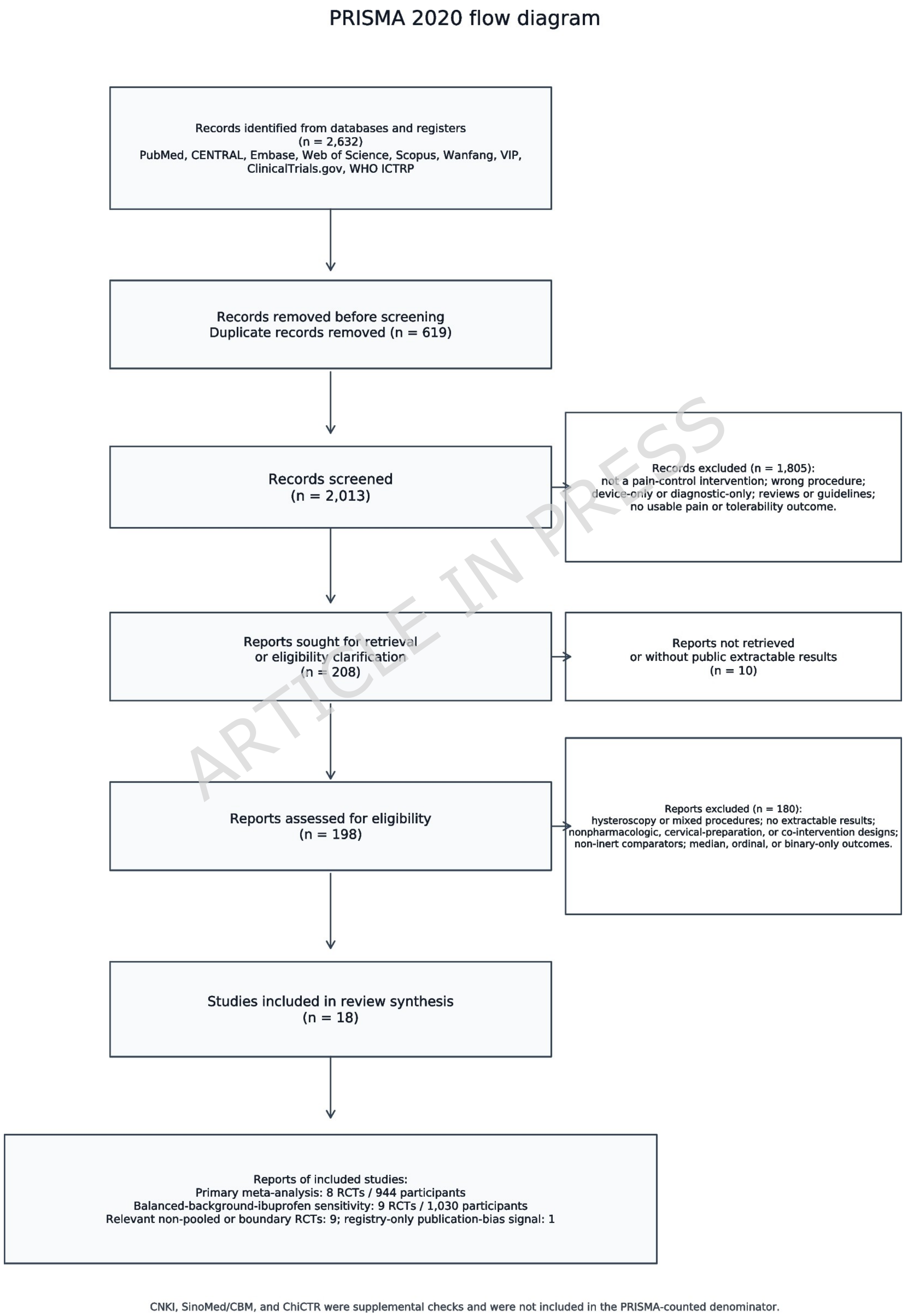

Figure 1. PRISMA 2020 flow diagram for study identification, deduplication, screening, retrieval, eligibility assessment, and inclusion, redrawn using the PRISMA 2020 template structure. Counts are based on the PRISMA-counted database/register search set described in the Methods, with documented exclusion and non-inclusion categories shown in the diagram and detailed in Supplementary Table S5.

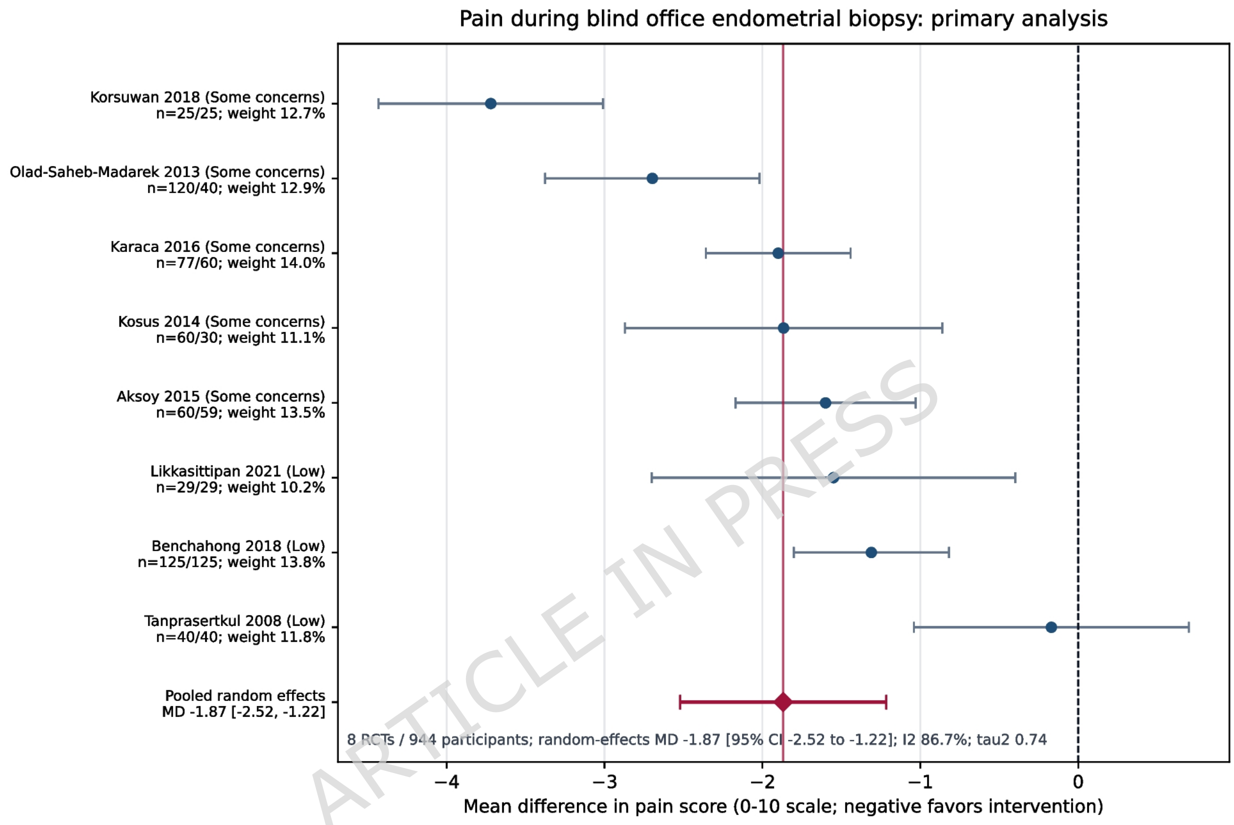

Figure 2. Primary DerSimonian-Laird random-effects meta-analysis of pain during blind office or outpatient endometrial biopsy. Negative mean differences favor active pain-control intervention. Study labels show the RoB 2 judgment, analyzed denominators, and random-effects weight. The red diamond and horizontal line show the pooled random-effects mean difference and 95% confidence interval. The analysis is restricted to placebo-, saline-, placebo-gel-, distilled-water-, or inert-comparator randomized trials with extractable intra-procedural pain scores on a 0-10 compatible scale. In the Olad-Saheb-Madarek 2013 row, n = 120/40 denotes the combined active local-anesthetic arms versus the distilled-water

placebo arm. Aksoy 2015 uses the outcome-level pain denominator of 60 versus 59 after one failed placebo-group procedure. Korsuwan 2018 uses the abstract/table denominator of 25 versus 25, with the text/table discrepancy retained in the extraction documentation.

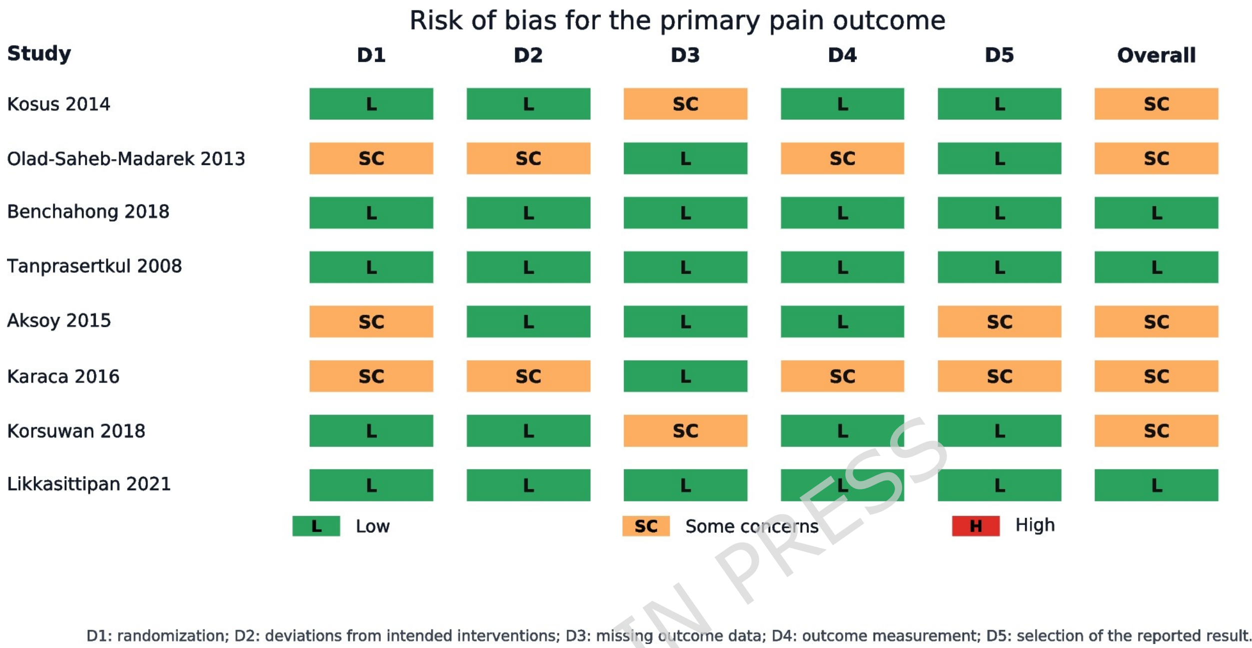

Figure 3. RoB 2 domain-level risk-of-bias judgments for the primary pain outcome. Domains are D1 randomization, D2 deviations from intended interventions, D3 missing outcome data, D4 outcome measurement, D5 selection of the reported result, and overall judgment. Green symbols indicate low risk of bias, orange symbols indicate some concerns, and the overall column gives the outcome-level judgment; no primary-analysis trial was judged at high risk of bias.

Tables

Table 1. Primary-analysis randomized trials, with country and intervention details.

Study (country)	n active/control	Intervention	Comparator	Primary timepoint	RoB 2
Kosus et al. 2014 (Turkey)	60/30	Intrauterine local anesthetic: levobupivacaine plus lidocaine	Saline	During biopsy	Some concerns

Study (country)	n active/control	Intervention	Comparator	Primary timepoint	RoB 2
Olad-Saheb-Madarek et al. 2013 (Iran)	120/40	Mixed local anesthetic: intrauterine lidocaine, cervical spray, or combined arms	Distilled-water placebo	During biopsy	Some concerns
Benchahong et al. 2018 (Thailand)	125/125	Intrauterine local anesthetic: lidocaine instillation	Normal saline	During biopsy	Low
Tanprasertkul et al. 2008 (Thailand)	40/40	Systemic COX-2 inhibitor: etoricoxib 120 mg	Placebo tablet	During biopsy	Low
Aksoy et al. 2015 (Turkey)	60/59	Cervical topical anesthetic: 10% lidocaine spray	Placebo spray	During biopsy	Some concerns
Karaca et al. 2016 (Turkey)	77/60	Cervical topical anesthetic: 2% lidocaine gel	Placebo gel	During biopsy	Some concerns
Korsuwan et al. 2018 (Thailand)	25/25	Cervical topical anesthetic: 10% lidocaine spray	Placebo/saline spray	During endometrial biopsy*	Some concerns
Likkasittipan et al. 2021 (Thailand)	29/29	Cervical topical anesthetic: 2% lidocaine gel	Placebo gel	During endometrial aspiration	Low

Note: *Korsuwan 2018 labelled the selected value as during endometrial biopsy in its table and figure, while the accompanying text used the broader phrase during the procedure. Likkasittipan 2021 explicitly reported pain during endometrial aspiration.

Table 2. Summary of Findings and outcome-level evidence profile.

Outcome	Participants/studies	Effect	Certainty	Explanation
Procedural pain during blind office/outpatient EMB or sampling	944 participants; 8 RCTs	MD -1.87 (95% CI -2.52 to -1.22) on a 0-10 scale	Low	Downgraded for RoB 2 some-concern judgments in 5/8 trials and serious inconsistency (I-squared = 86.7%), reflecting substantial heterogeneity and clinically different intervention mechanisms. Publication bias remained unresolved because the number of trials was small and one registry record had no posted results. The direction of that unavailable result was unknown, and small-study effects could not be assessed reliably; publication bias was therefore not judged strongly suspected enough to warrant an additional downgrade. This domain judgment remains uncertain.
Serious adverse events	Sparse/inconsistently reported	No clear serious-harm signal identified	Very low	Downgraded for adverse-event ascertainment concerns, sparse events and small trials, inconsistent definitions/follow-up windows, incomplete arm-level reporting, and unresolved reporting bias; quantitative pooling was not appropriate.

Table 3. Prior-review crosswalk and non-pooled or sensitivity evidence.

Study	Why it matters	Prespecified boundary category	Handling in this review
Sripha 2018	Placebo-controlled lidocaine-spray office EMB trial	Median/range only	Non-pooled; median/range only.
Luangtangvarodom 2018	Placebo-controlled lidocaine-spray Novak sampling trial	Binary severe-pain threshold only	Non-pooled; binary severe-pain threshold only, no arm-level mean/SD.
Piyawetchakarn 2019	Three-arm lidocaine-spray/placebo/no-intervention Pipelle trial	Median/range only; three-arm boundary	Non-pooled; median/range biopsy pain only, no mean/SD conversion.
Chayuttawanichakul 2022	Placebo-controlled lidocaine-gel endometrial-sampling trial	Median-only outcome reporting	Non-pooled; median-only procedural pain summaries.
Hui 2006	Randomized double-blind intrauterine lignocaine trial	Original extractable source data unavailable; median-only secondary source	Non-pooled; median VAS data and no extractable arm-level mean/SD in available source.
Siddle 1983	Historical double-blind naproxen/placebo Vabra suction-curettage trial	Historical Vabra/suction-curettage boundary	Non-pooled; no arm-level during-procedure mean/SD in indexed abstract.
Trolice 2000	Randomized double-masked intrauterine lidocaine/saline Pipelle trial	Median/range only; 20-cm VAS	Non-pooled; median/range on 20-cm VAS, incompatible with the prespecified mean/SD pool.
Kozman 2001	Randomized double-blind lignocaine-gel/inert-gel Vabra sampling trial	Vabra sampling; binary high-pain threshold	Non-pooled; binary high-pain result, not continuous mean/SD primary endpoint.
Kaya 2015	Placebo/intrauterine lidocaine with or without rectal diclofenac office EMB trial	Co-intervention design	Boundary evidence; co-intervention and extractability do not fit the primary pool.
Tueman 2026	Placebo-gel trial with ibuprofen in both groups	Balanced-background ibuprofen sensitivity	Included only in balanced-background-ibuprofen sensitivity analysis.
NCT02465320	Completed placebo-controlled COL-1077 registry record, 187 participants, no posted results	Registry-only record without posted results	Used for qualitative publication-bias discussion.